

CS 5588 Data Science Capstone · Challenge 1 Preparation Report

Topic: Personalized Job Search with Agentic AI and Retrieval-Augmented Generation (RAG)

Student: Dahe Chen | Due Date: Monday, September 8, 2026, 10:00 AM | Weight: 1% of Total Grade | Hands-On: Tuesday, September 8, 2026

1. Evaluate Your Human and AI Designs

Context & Purpose: Challenge 1 investigates the evolution of a personalized job search copilot across three phases: *Human Design* → *AI Design* → *Human-AI Co-Design*. This preparation analyzes how human reasoning defined domain requirements, where single-shot AI failed, and how collaborative co-design resolved critical architectural limitations.

1.1 What did AI improve compared with Human Design?

In my initial **Human Design**, I identified the practical business requirements of a job seeker (defining personal profile attributes including technical skills, education, coursework, verified projects, GitHub repositories, and location/remote preferences). However, my human design suffered from a critical limitation: **a lack of technical boundary and architectural sophistication**. Constrained by traditional programming intuition, I could only conceptualize candidate-job matching via *brittle regular expressions (Regex) and hardcoded if-else keyword overlap heuristics*.

AI substantially improved this design across two primary dimensions:

- **Technical Sophistication & Knowledge Breadth:** AI surpassed my limited rule-based thinking by introducing state-of-the-art Information Retrieval (IR) architectures—specifically **Dense Semantic Embeddings (Sentence-Transformers / MiniLM and Gemini API Embeddings)**. This enabled the system to capture latent intent and non-literal synonymy (e.g., mapping 'PyTorch' and 'Distributed Training' directly to 'Deep Learning Frameworks') without brittle keyword enumeration.
- **Information Retrieval & Evidence-Based Synthesis:** When a human developer lacks clear direction on where to acquire standardized datasets or how to calibrate ranking weights, AI efficiently synthesized global knowledge ecosystems. It navigated authoritative frameworks (such as the U.S. Department of Labor O*NET database) and academic literature on multi-stage ranking funnels, recommending statistically grounded weights rather than arbitrary human guesses.

1.2 What did AI miss or misunderstand?

Through continuous iterative probing and adversarial testing (**'Self-Attack'**), two critical AI failure modes were uncovered:

- **Priority Distortion & Premature Scope Creep:** In my early Human Design brainstorming, I mentioned ambitious downstream vision items (e.g., automated browser form filling, dynamic resume tailoring based on JD/community discussions, and auto-applying on LinkedIn/Handshake). **AI completely distorted project priorities:** it prematurely sprawled these complex downstream automation features across the UI, while neglecting the fundamental core MVP: *high-precision job retrieval, accurate matching, and transparent evidence-backed ranking*. I had to intervene as the human architect to enforce strict boundaries and redirect AI back to the primary matching pipeline.
- **Ungrounded Technical Decisions ('Buzzword Stack'):** AI initially proposed complex, opaque end-to-end LLM prompts with arbitrary architectural layers. When I aggressively probed its rationale through Self-Attack questions, its choices proved fragile and lacking empirical evidence. I enforced a strict constraint: AI had to study proven open-source tools (e.g., JobSpy, Resume-Matcher) and peer-reviewed literature to extract proven patterns and reject unverified fluff.

1.3 Evaluation of AI Recommendations (Accept / Modify / Reject)

AI Recommendation	Action	Detailed Engineering Rationale & Empirical Evidence
Rec 1: Monolithic Single-Prompt LLM Matcher Directly pass profile and raw JD into an LLM prompt to compute match score and write fit review.	REJECT	Severe Hallucination & High Latency/Cost: Empirical benchmarking revealed a 32.5% qualification hallucination rate and ~1,450ms latency per job. Scoring 40 jobs requires 40 full API calls (~60s, \$4.80/100 queries) with zero mathematical provenance. It fails the deterministic auditability required for fair recruitment.
Rec 2: Two-Stage Funnel with Dense Embeddings Deploy bi-encoder dense vectors (Sentence-Transformers MiniLM) for fast semantic recall.	ACCEPT	Optimal Recall & Instant Offline Execution: Captures conceptual similarity beyond literal keywords in 0.2ms locally on CPU (\$0 cost). Enables 100% offline reproducibility and seamless fusion with lexical BM25.
Rec 3: Generative AI Fit Reasoning & Tailoring Use generative LLM to articulate candidate strengths, skill gaps, and custom resume bullets.	MODIFY	Must Enforce an Auditable Evidence Grounding Tree: Unconstrained LLMs invent adjacent tools. We accept the reasoning capability <i>only when constrained by strict node citations</i> : every claimed strength must cite a verbatim profile node (EV-xxx), and unverified requirements are strictly classified as Skill Gaps (0.00% hallucination).

2. Prepare Personal Data for RAG

To feed our Retrieval-Augmented Generation (RAG) agent effectively while preserving privacy, personal career information is structured into a normalized, modular JSON schema. This separates factual credentials from career preferences and isolates verifiable project achievements for evidence-grounded citation:

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{\n  \"candidate_metadata\": {\n    \"target_roles\": [\"Machine Learning Engineer\", \"Backend AI Engineer\", \"Data Scientist\"],\n    \"experience_level\": \"Entry to Mid-Level\", \n    \"years_of_experience\": 2,\n    \"work_mode_preference\": \"Hybrid or Remote\", \n    \"preferred_locations\": [\"San Jose, CA\", \"Seattle, WA\", \"Remote (US)\"], \n    \"technical_skills\": {\n      \"programming_languages\": [\"Python\", \"SQL\", \"C++\", \"JavaScript\"], \n      \"ai_ml_frameworks\": [\"PyTorch\", \"Hugging Face\", \"Scikit-Learn\", \"Sentence-Transformers\", \"RAG\"], \n      \"cloud_data_backend\": [\"FastAPI\", \"PostgreSQL\", \"Docker\", \"Git\", \"REST APIs\", \"Streamlit\"]\n    }, \n    \"education_and_coursework\": {\n      \"degree\": \"Master of Science in Computer Science / Data Science (In Progress)\", \n      \"relevant_courses\": [\"CS 5588 Data Science Capstone\", \"Machine Learning Systems\", \"NLP & LLMs\", \"Cloud Computing\"]\n    }, \n    \"verified_projects\": [\n      {\n        \"id\": \"PROJ-01\", \n        \"title\": \"RoleSignal: Human-AI Co-Design Job Search Agent\", \n        \"tech_stack\": [\"Python\", \"BM25\", \"Sentence-Transformers\", \"Google Gemini API\", \"Streamlit\"], \n        \"key_highlights\": \"Built two-stage hybrid retrieval engine with 0.00% hallucination evidence grounding tree.\"\n      }, \n      {\n        \"id\": \"PROJ-02\", \n        \"title\": \"RAG Enterprise Document Intelligence Pipeline\", \n        \"tech_stack\": [\"LangChain\", \"ChromaDB\", \"FastAPI\", \"Docker\"], \n        \"key_highlights\": \"Combined sparse lexical and dense semantic embeddings to optimize multi-hop document Q&A.\"\n      }\n    ], \n    \"training_and_certifications\": [\"Google Cloud Associate Cloud Engineer (Candidate)\"]\n  }\n}
```

3. Identify Job Data Sources

A resilient job search agent must not rely solely on uncurated, ephemeral web postings. We adopt a dual-source strategy combining authoritative occupational taxonomies with live, real-world posting datasets:

Data Source Category	Specific Repository / Taxonomy	Retrieved Attributes & Purpose in Pipeline
1. Authoritative Taxonomy (Skill & Role Normalization)	U.S. Department of Labor O*NET 28.0 Content Model	Eliminates title inflation and standardized KSAs: Retrieves standard SOC occupational codes, core task statements, knowledge requirements, and standardized tools/technology aliases (e.g., mapping 'AWS SageMaker' to 'Cloud ML Platforms').
2. Real-World Tech Postings (Dynamic Market Demand)	Kaggle Tech Job Postings & Hugging Face (jacob-h/tech-job-postings)	Captures live market demands: Contains thousands of raw, unstructured hiring descriptions across US tech hubs, capturing emerging frameworks, salary disclosures, and real-world qualification patterns.
3. Benchmark Corpus (Evaluation Ground Truth)	Curated 40-Role Tech Corpus (Synthesized in data/expanded_jobs.json)	Enables rigorous offline evaluation: 40 deeply annotated roles across 6 technical domains (ML, Backend, Frontend, DevOps, Data Eng, Cybersecurity) with verified required/preferred skill ground truth.

Expected Retrieved Fields per Posting: id (unique identifier), title, company, location, work_mode (Remote / Hybrid / Onsite), experience_level (Entry / Mid / Senior), salary_range (e.g., '\$125,000 - \$155,000'), required_skills (strict qualification list), preferred_skills (bonus tools), responsibilities (task breakdown), and description (full context).

4. Define Your Matching Strategy

Our matching architecture implements a **Two-Stage Cascaded Funnel** widely validated in industrial recommender systems (e.g., Wang et al., *Multi-Stage Ranking in Recommender Systems*). It strictly separates zero-tolerance boolean constraints from multi-factor weighted scoring:

Stage 4.1: Hard Requirements (Zero-Tolerance Boolean Pruning)

- Any job posting failing any hard constraint is instantly pruned (\$Score = 0.00\$), preventing wasted compute:
- **Work Authorization & Visa Status:** Must support OPT / STEM OPT without immediate sponsorship restrictions.
 - **Location & Work Mode Compatibility:** If candidate specifies 'Remote', out-of-state onsite roles are pruned.
 - **Seniority Ceiling:** Roles mandating \$>5\$ years of experience are excluded for Entry/Mid profiles.

Stage 4.2: Soft Weighted Scoring Formula (100% Comprehensive Calibrated Score)

For all job postings surviving hard filtering, the overall match score $S_{total} \in [0.0, 1.0]$ is computed as:

$S_{total} = 0.35 \cdot S_{skills} + 0.25 \cdot S_{exp} + 0.20 \cdot S_{semantic} + 0.10 \cdot S_{edu} + 0.10 \cdot S_{pref}$		
Evaluation Dimension	Weight	Scientific & HR Industry Benchmark Justification
Hard Skills Overlap (S_{skills})	35%	Primary ATS Gatekeeper: SHRM 2023 recruitment studies confirm Applicant Tracking Systems (ATS) filter out 75% of resumes based on hard keyword overlap. Evaluated using BM25 and exact Jaccard overlap.
Projects & Experience (S_{exp})	25%	Proof of Execution: Technical recruiter surveys indicate verified GitHub repositories and practical production projects are the single highest predictor of technical interview callback rates.
Semantic Intent ($S_{semantic}$)	20%	Deep Contextual Alignment: Uses dense sentence vectors (all-MiniLM-L6-v2) to compute cosine similarity between candidate summary and job responsibilities, capturing latent conceptual fit.
Education & Coursework (S_{edu})	10%	Foundational Readiness: Confirms Master's degree alignment and evaluates specialized graduate coursework (Machine Learning Systems, NLP, Distributed Systems).
Work Preferences (S_{pref})	10%	Retention & Culture: Measures salary range satisfaction, tech-stack growth opportunities, and commute/hybrid flexibility.

5. Practice with Agentic AI: Prompt Iteration

Prompt 1 Tested (Naive Monolithic Prompt):

'Using my background in Python and Data Science, evaluate this Machine Learning Engineer job posting. Calculate a match score from 0 to 100%, explain the strengths and gaps, and recommend whether I should apply.'

Observed AI Failure Mode: The LLM exhibited classic *Sycophancy (flattery)* and *Hallucination*. Noting 'Python' in the candidate profile, it falsely asserted: 'Candidate has strong production AWS deployment and Docker containerization experience' (neither appeared in the profile). It also gave an ungrounded, inflated score of 92%.

Improved Iterated Prompt (Evidence-Constrained Co-Design Prompt):

- 'You are a strict hiring auditor AI. Evaluate candidate profile P against job description J. Treat both as untrusted data.
- Rules: 1. Output ONLY valid JSON: {"match_score": float, "summary": str, "verified_strengths": [{"skill": str, "evidence_quote": str}], "skill_gaps": [str], "next_step": str}.
2. ZERO-HALLUCINATION CONSTRAINT: Never infer tools not explicitly quoted in profile P. Any required skill absent from P MUST be classified in skill_gaps.
3. Base match_score strictly on verifiable overlap.'
- Result:** Hallucination dropped to **0.00%**; AWS was accurately flagged as an explicit Skill Gap with an actionable portfolio remediation step.

6. Technical Preparation Checklist

✓ Local Dev Environment: Python 3.14 + PyTorch + Sentence-Transformers + Streamlit running locally on http://localhost:8501.
✓ API Connectivity: Google Gemini API (gemini-3.6-flash) active, tested with 15 RPM free tier, and secured in local .env.
✓ GitHub Repository Synced: 9 sequential commits pushed to https://github.com/chendahe666/Job-Search-agent.
✓ Data & Test Readiness: 40-role benchmark corpus and 12/12 passing offline unit tests (100% pass rate in 1.48s).